

1 Formulas Used in This Topic

1.1 Expected Value of a Discrete Random Variable

$$E[X] = \sum_{i=1}^n x_i p_i$$

Where

- X random variable representing uncertain outcomes
- x_i the i -th possible outcome of X
- p_i probability associated with outcome x_i
- n number of possible outcomes

Context Represents the long-run average outcome under repeated identical conditions.

1.2 Variance of a Discrete Random Variable

$$\text{Var}(X) = \sum_{i=1}^n (x_i - E[X])^2 p_i$$

Where

- x_i possible outcome
- $E[X]$ expected value of the distribution
- p_i probability of outcome x_i
- n number of outcomes

Context Measures dispersion and quantifies risk or volatility.

1.3 Expected Monetary Value (EMV)

$$EMV(A_j) = \sum_{i=1}^m p_i \text{Payoff}(A_j, S_i)$$

Where

- A_j decision alternative (for example a strategy)
- S_i state of nature (future scenario)
- $\text{Payoff}(A_j, S_i)$ monetary result of A_j under state S_i

- p_i probability of state S_i
- m number of states of nature

Context Represents the average monetary performance of a strategy under uncertainty.

1.4 Maximin Criterion

$$\text{Maximin}(A_j) = \max_j \left(\min_i \text{Payoff}(A_j, S_i) \right)$$

Where

- A_j decision alternatives
- S_i states of nature
- $\min_i \text{Payoff}(A_j, S_i)$ worst-case payoff of A_j
- outer max selects the highest worst-case result

Context Represents a conservative decision maker seeking safety.

1.5 Maximax Criterion

$$\text{Maximax}(A_j) = \max_j \left(\max_i \text{Payoff}(A_j, S_i) \right)$$

Where

- inner max best payoff each alternative can achieve
- outer max selects the alternative with the largest best-case payoff

Context Represents an optimistic decision maker seeking maximum possible gain.

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